

CMPINF 0010 Final Project PLEM Results

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Introduction:

The biggest question that was given to us for our project was, “What is the best neighborhood in Pittsburgh?” Using datasets from the Western Pennsylvania Regional Data Center, we created a metric called the **Pittsburgh Livability & Engagement Metric (PLEM)**, which evaluates different neighborhoods around Pittsburgh based on activity, engagement, and economic stability. Each team member analyzed one dataset and created a sub-metric. These sub-metrics were then normalized and combined to determine the final PLEM score for each neighborhood.

The PLEM Metric:

The Metric measured how well a neighborhood supports an active, stable, and connected community using these 3 components:

Sub-Metric	Dataset	What it Measures	Contributor
Parking Transactions → Activity & Accessibility	Parking Transactions	Daily vibrancy, parking demand, commercial activity	Jacob
Community Center Attendance → Community Engagement	Daily Community Center Attendance	Social participation and engagement	Piotr
Licensed Businesses / Trades → Economic Stability	Licensed Businesses, Contractors & Trades	Business longevity, service availability	Chase

Final PLEM Score = Normalized(Activity + Engagement + Stability)

Sub-Matric Results:

1. Parking Transactions — Activity & Vibrancy

Top indicators:

- High daily transaction volume
- Consistent parking demand
- Commercial and visitor activity

Top Neighborhoods:

- Most Vibrant: Oakland, Downtown
- Moderate: Shadyside, Lawrenceville
- Least Vibrant: East Liberty, Squirrel Hill

2. Community Center Attendance — Engagement

Top indicators:

- Average daily attendance
- Year-over-year growth
- Annual community participation

Top Neighborhoods:

1. West Penn Community Center (North Oakland / Oakland)
2. Warrington Community Center (South Side)
3. Phillips Community Center (Carrick)

3. Economic Stability — Licensed Businesses & Trades

Top indicators:

- Average years active
- Total licensed businesses
- Service and building coverage

Top Neighborhoods:

1. Downtown
2. Oakland
3. South Side
4. Lawrenceville

Final Combined Results - PLEM Rankings

After normalizing all sub-matrices and combining them, we were able to produce a final list ranking of neighborhoods based on our data.

Best Overall Neighborhood: Oakland

- Ranks
 - #1 Vibrancy
 - Most active parking, meaning a lot of people coming in and out on a daily basis
 - #1 Community Center (West Penn Community Center)
 - Highest average annual attendance
 - Consistent Year over Year growth
 - #2 Economic Stability
 - High number of active businesses
 - Long business lifespans

Conclusion

Based on our Pittsburgh Livability & Engagement Metric (PLEM), Oakland is the best neighborhood in Pittsburgh. It is the only neighborhood that performed near the top in all three categories:

- Most vibrant
- Most engaged
- Economically stable

Oakland's combination of activity, community participation, and long-standing businesses makes it the strongest overall neighborhood according to our data-driven analysis.